

CSE 400: Fundamentals of Probability in Computing

Tutorial 2 – Solutions

Q1. Random Point on a Line Segment

Problem Interpretation

A point is chosen uniformly at random on a line segment of length L .

Let:

$$x \in (0, L)$$

Two segments formed: x and $L - x$

Define the ratio:

$$R = \frac{\max(x, L - x)}{\min(x, L - x)}$$

We need:

$$P\left(R < \frac{1}{4}\right)$$

Step-by-Step Reasoning

Case 1: $x \leq \frac{L}{2}$

Shorter segment = x

Longer segment = $L - x$

$$\frac{x}{L - x} < \frac{1}{4}$$

$$4x < L - x \Rightarrow 5x < L \Rightarrow x < \frac{L}{5}$$

Case 2: $x > \frac{L}{2}$

Shorter segment = $L - x$

Longer segment = x

$$\frac{L - x}{x} < \frac{1}{4}$$

$$4L - 4x < x \Rightarrow 5x > 4L \Rightarrow x > \frac{4L}{5}$$

Valid Region

$$(0, \frac{L}{5}) \cup (\frac{4L}{5}, L)$$

Total valid length:

$$\frac{L}{5} + \frac{L}{5} = \frac{2L}{5}$$

Final Probability

$$P = \frac{2L/5}{L} = \frac{2}{5}$$

Q2. Bayesian Classification with Normal Distributions

Given

$$X|W \sim N(4, 4), \quad X|B \sim N(6, 9)$$

$$P(B) = \alpha, \quad P(W) = 1 - \alpha$$

Observed:

$$X = 5$$

Condition:

$$P(B|X = 5) = P(W|X = 5) = \frac{1}{2}$$

Applying Bayes' Theorem

$$\alpha f(5|B) = (1 - \alpha) f(5|W)$$

Normal Density Values

$$f(5|B) = \frac{1}{3\sqrt{2\pi}} e^{-1/18}, \quad f(5|W) = \frac{1}{2\sqrt{2\pi}} e^{-1/8}$$

Substitution

$$\frac{\alpha e^{-1/18}}{3\sqrt{2\pi}} = \frac{(1 - \alpha) e^{-1/8}}{2\sqrt{2\pi}}$$

Simplifies to:

$$\frac{3\alpha}{3\alpha + 2(1 - \alpha)e^{-5/72}} = \frac{1}{2}$$

Solving

$$6\alpha = 3\alpha + 2(1 - \alpha)(0.93)$$

$$3\alpha = 1.86 - 1.86\alpha \Rightarrow 4.86\alpha = 1.86 \Rightarrow \alpha \approx 0.3827$$

Final Answer

$$\alpha \approx 0.38$$

Q3. Optimal Location Problem

(a) Uniform Distribution

$$X \sim \text{Uniform}(0, A), \quad f_X(x) = \frac{1}{A}$$

$$E[|X - a|] = \int_0^A |x - a| \frac{1}{A} dx$$

Breaking absolute value:

$$E[|X - a|] = \int_0^a (a - x) \frac{1}{A} dx + \int_a^A (x - a) \frac{1}{A} dx$$

After simplification:

$$\frac{d}{da} \left(\frac{2a^2 - 2aA + A^2}{2A} \right) = \frac{4a - 2A}{2A}$$

Minimization:

$$4a - 2A = 0 \Rightarrow a = \frac{A}{2}$$

Conclusion: midpoint.

(b) Exponential Distribution

$$X \sim \text{Exponential}(\lambda), \quad f_X(x) = \lambda e^{-\lambda x}$$

$$E[|X - a|] = \int_0^a (a - x) \lambda e^{-\lambda x} dx + \int_a^\infty (x - a) \lambda e^{-\lambda x} dx$$

Derivative:

$$\frac{d}{da} E[|X - a|] = 1 - 2e^{-\lambda a}$$

Minimization:

$$1 - 2e^{-\lambda a} = 0 \Rightarrow e^{-\lambda a} = \frac{1}{2} \Rightarrow a = \frac{\ln 2}{\lambda}$$

Q4. Mixture of Exponentials

$$P(T > t) = p_1 e^{-\lambda_1 t} + p_2 e^{-\lambda_2 t}$$

$$P(T > t + s | T > t) = \frac{p_1 e^{-\lambda_1(t+s)} + p_2 e^{-\lambda_2(t+s)}}{p_1 e^{-\lambda_1 t} + p_2 e^{-\lambda_2 t}}$$

Q5. Conditional Distribution of Poisson Variables

$$X \sim \text{Poisson}(\lambda_1), \quad Y \sim \text{Poisson}(\lambda_2)$$

$$P(X = k | X + Y = n) = \frac{P(X = k, Y = n - k)}{P(X + Y = n)}$$

After simplification:

$$\binom{n}{k} \left(\frac{\lambda_1}{\lambda_1 + \lambda_2} \right)^k \left(\frac{\lambda_2}{\lambda_1 + \lambda_2} \right)^{n-k}$$

Q6. Transformation of Random Variable

$$X \sim \text{Uniform}[-2, 2]$$

$$F_Y(y) = P(Y \leq y)$$

Piecewise:

$$F_Y(y) = \begin{cases} 0 & y < -2 \\ \frac{y+2}{4} & -2 \leq y < -1 \\ \frac{1}{4} & -1 \leq y < 0 \\ \frac{1}{2} & y = 0 \\ \frac{3}{4} & 0 < y < 1 \\ \frac{y+2}{4} & 1 \leq y \leq 2 \\ 1 & y > 2 \end{cases}$$
$$f_Y(y) = \begin{cases} \frac{1}{4} & -2 < y < -1 \\ \frac{1}{4} & 1 < y < 2 \\ 0 & \text{otherwise} \end{cases}$$

Q7. Stock Model Approximation

$$S_{1000} = S_0 u^k d^{1000-k}$$

Condition:

$$S_{1000} \geq 1.30 S_0$$

Log transformation:

$$k \ln u + (1000 - k) \ln d \geq \ln(1.30)$$

$$k \geq \frac{\ln(1.30) - 1000 \ln d}{\ln u - \ln d} \approx 469.4 \Rightarrow k \geq 470$$

$$k \sim \text{Binomial}(1000, 0.52)$$

$$P(k \geq 470) = \sum_{k=470}^{1000} \binom{1000}{k} (0.52)^k (0.48)^{1000-k}$$

Q8. Exponential Distribution

$$\lambda = \frac{1}{2}$$

(a)

$$P(X > 2) = e^{-1}$$

(b)

$$P(X \geq 10 | X > 9) = e^{-1/2}$$

Q9. Distribution of $Z = X^2 + Y^2$

$$F_Z(z) = P(X^2 + Y^2 \leq z)$$

$$F_Z(z) = \int_{-z}^z \int_{-\sqrt{z^2-x^2}}^{\sqrt{z^2-x^2}} f_X(x) f_Y(y) dy dx$$

$$f_Z(z) = \frac{d}{dz} F_Z(z)$$

Using Leibniz rule:

$$f_Z(z) = z \int_{-z}^z f_X(x) \left[f_Y(\sqrt{z^2-x^2}) + f_Y(-\sqrt{z^2-x^2}) \right] dx$$

Q10. Joint Distribution of Maxima

$$M_n = \max(X_1, \dots, X_n)$$

$$P(M_n \leq x) = F(x)^n$$

$$M_{n+1} = \max(M_n, X_{n+1})$$

$$P(M_n \leq x, M_{n+1} \leq y) = \begin{cases} F(x)^n F(y) & x \leq y \\ F(y)^{n+1} & x > y \end{cases}$$